

35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — FEET (1 foot = 12 inches)							
FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND FEET PER INCH (x in ratio) Use ratio below or $\frac{\text{SCALE}}{12}$	FEET PER INCH	TOTAL FEET ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (y in ratio) Use ratio below or $\frac{\text{Feet on scale}}{\text{Feet per inch}}$	TOTAL SCALE LENGTH (INCHES)
	Scale Unit :	represents Map Unit					
1:12 000	1 inch	12 000 in	$\frac{12}{1} = \frac{12\ 000}{x}$	1000.000	6000	$\frac{1000.000}{1} = \frac{6000}{y}$	6.000
1:20 000	1 inch	20 000 in	$\frac{12}{1} = \frac{20\ 000}{x}$	1666.6666	8000	$\frac{1666.6666}{1} = \frac{8000}{y}$	4.800
1:24 000	1 inch	24 000 in	$\frac{12}{1} = \frac{24\ 000}{x}$	2000.000	8000	$\frac{2000.000}{1} = \frac{8000}{y}$	4.000
1:25 000	1 inch	25 000 in	$\frac{12}{1} = \frac{25\ 000}{x}$	2083.3333	8000	$\frac{2083.3333}{1} = \frac{8000}{y}$	3.840
1:31 250	1 inch	31 250 in	$\frac{12}{1} = \frac{31\ 250}{x}$	2604.1666	8000	$\frac{2604.1666}{1} = \frac{8000}{y}$	3.072
1:31 680	1 inch	31 680 in	$\frac{12}{1} = \frac{31\ 680}{x}$	2640.000	8000	$\frac{2640.000}{1} = \frac{8000}{y}$	3.030
1:48 000	1 inch	48 000 in	$\frac{12}{1} = \frac{48\ 000}{x}$	4000.000	24 000	$\frac{4000.000}{1} = \frac{24\ 000}{y}$	6.000
1:50 000	1 inch	50 000 in	$\frac{12}{1} = \frac{50\ 000}{x}$	4166.6666	24 000	$\frac{4166.6666}{1} = \frac{24\ 000}{y}$	5.760
1:62 500	1 inch	62 500 in	$\frac{12}{1} = \frac{62\ 500}{x}$	5208.3333	30 000	$\frac{5208.3333}{1} = \frac{30\ 000}{y}$	5.760
1:63 360	1 inch	63 360 in	$\frac{12}{1} = \frac{63\ 360}{x}$	5280.000	30 000	$\frac{5280.000}{1} = \frac{30\ 000}{y}$	5.681
1:72 000	1 inch	72 000 in	$\frac{12}{1} = \frac{72\ 000}{x}$	6000.000	30 000	$\frac{6000.000}{1} = \frac{30\ 000}{y}$	5.000
1:75 000	1 inch	75 000 in	$\frac{12}{1} = \frac{75\ 000}{x}$	6250.000	30 000	$\frac{6250.000}{1} = \frac{30\ 000}{y}$	4.800
1:96 000	1 inch	96 000 in	$\frac{12}{1} = \frac{96\ 000}{x}$	8000.000	33 000	$\frac{8000.000}{1} = \frac{33\ 000}{y}$	4.125
To find feet per inch on 1: 12 000 map . . . 12 inches = 1 foot Show in ratio as ... $\frac{12}{1}$ inches feet Let SCALE (12 000) be in inches Fractional scale says 1 inch represents 12,000 in Let x be feet that 1 inch represents on map Show in ratio as ... $\frac{12\ 000}{x}$ inches feet Solution . . . $\frac{12}{1} = \frac{12\ 000}{x}$ $12 \cdot x = 12\ 000 \cdot 1$ $\frac{12\ x = 12\ 000}{12 \quad 12}$ $x = \frac{12\ 000}{12} \text{ (SCALE)}$ $x = 1000.00$							